

Building a Site-to-Site VPN in AWS

Introduction

In real-world enterprise environments, companies often need to connect their **on-premises data centers** to their **AWS Cloud VPCs** securely.

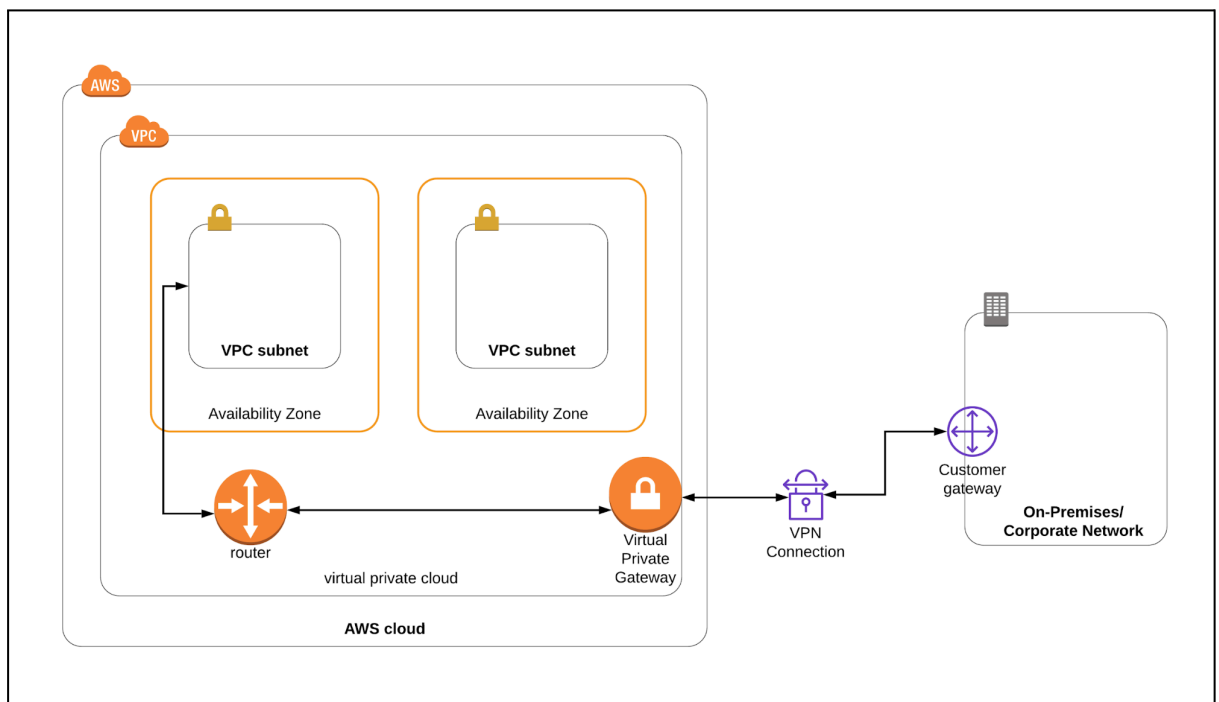
AWS Site-to-Site VPN provides a reliable and encrypted tunnel using the IPsec protocol, enabling private communication between these two networks.

But what if you don't have a physical on-premises network? In this lab, we will **simulate an on-premises environment using a second VPC in the same AWS region (us-east-1)**. One VPC will represent the **corporate/on-premises side**, while the other will act as the **AWS Cloud network**. We will then connect them using a Site-to-Site VPN.

What is AWS Site-to-Site VPN?

A VPN connection refers to the connection between your VPC and your own on-premises network. Site-to-Site VPN supports Internet Protocol security (IPsec) VPN connections.

By default, instances that you launch into an Amazon VPC can't communicate with your own (remote/on-premises) network.



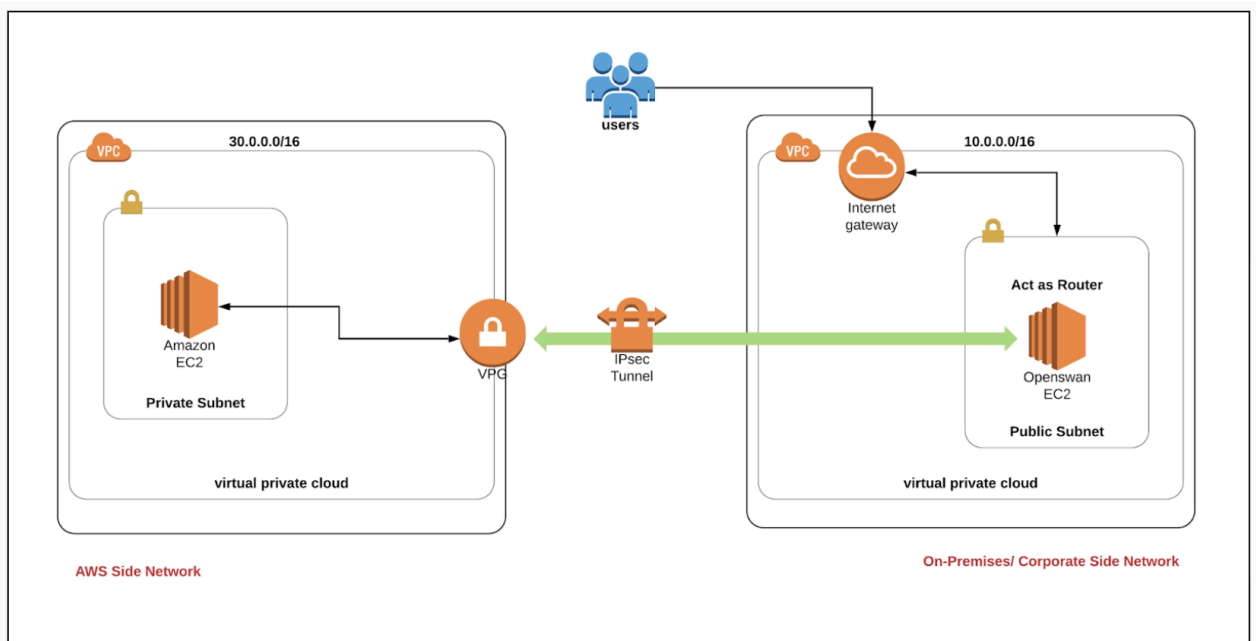
- **VPN connection:** A secure connection between your on-premises equipment and your VPCs.
- **VPN tunnel:** An encrypted link where data can pass from the customer network to or from AWS.
- **Customer gateway:** An AWS resource which provides information to AWS about your customer gateway device.
- **Customer gateway device:** A physical device or software application on your side of the Site-to-Site VPN connection.
- **Virtual private gateway:** The VPN concentrator on the Amazon side of the Site-to-Site VPN connection. You use a virtual private gateway or a transit gateway as the gateway for the Amazon side of the Site-to-Site VPN connection.

Architecture Overview

Our lab consists of two VPCs in **N. Virginia (us-east-1)**:

- **VPC-OnPrem (Corporate Simulation)**
 - Public subnet
 - EC2 instance with **Openswan** (acting as a customer router)
- **VPC-AWS (AWS Cloud Network)**
 - Private subnet
 - EC2 instance to test private connectivity
- **VPN Connection:**
 - **Customer Gateway (CGW)** on AWS side (logical representation of the on-prem router)
 - **Virtual Private Gateway (VGW)** on AWS side (VPN concentrator)
 - **Two VPN tunnels** (AWS always provides redundancy, though in this lab we will focus on one active tunnel)

 *Tip: Always make sure your VPC CIDR blocks do not overlap; otherwise, routing will fail.*



Step 1: Create the On-Premises Simulation VPC.

1. Go to **VPC > Create VPC**
2. Navigate to **Your VPC's** on the left panel and click on the **Create VPC**.
 - Select **VPC only**
 - Name tag : Enter **On_Premises_Network**
 - IPv4 CIDR block : Enter **10.0.0.0/16**

Create VPC [Info](#)

A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances.

VPC settings

Resources to create [Info](#)
Create only the VPC resource or the VPC and other networking resources.

☒ VPC only ☐ VPC and more

Name tag - optional
Creates a tag with a key of 'Name' and a value that you specify.

On_Premises_Network

IPv4 CIDR block [Info](#)
☒ IPv4 CIDR manual input
☐ IPAM-allocated IPv4 CIDR block

IPv4 CIDR
10.0.0.0/16
CIDR block size must be between /16 and /28.

IPv6 CIDR block [Info](#)
☒ No IPv6 CIDR block
☐ IPAM-allocated IPv6 CIDR block
☐ Amazon-provided IPv6 CIDR block
☐ IPv6 CIDR owned by me

Tenancy [Info](#)
Default

Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key **Value - optional**

Q Name X Q On_Premises_Network X [Remove tag](#)

[Add tag](#)

You can add 49 more tags

[Cancel](#) [Preview code](#) [Create VPC](#)

3. Leave everything else as default and click on the **Create VPC**.

Step 2: Create a Public subnet

1. Subnets → Create Subnet.

- VPC ID : Select the **On_Premises_Network** VPC from the list.
- Subnet name : Enter **Public_subnet**
- Availability Zone :No preference
- IPv4 CIDR block : Enter **10.0.1.0/24**

The screenshot shows the 'Create subnet' page in the AWS Management Console. The breadcrumb navigation at the top indicates the path: VPC > Subnets > Create subnet. The page is divided into two main sections: 'VPC' and 'Subnet settings'.

VPC Section:

- VPC ID:** A dropdown menu showing 'vpc-04d4403d67a7a052e (On_Premises_Network)'.
- Associated VPC CIDRs:** A table showing 'IPv4 CIDRs' with the value '10.0.0.0/16'.

Subnet settings Section:

Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

- Subnet name:** A text input field containing 'Public_subnet'.
- Availability Zone:** A dropdown menu with 'No preference' selected.
- IPv4 VPC CIDR block:** A dropdown menu with '10.0.0.0/16' selected.
- IPv4 subnet CIDR block:** A text input field with '10.0.1.0/24' and a '256 IP' indicator.

Tags - optional

Key: Name | Value - optional: Public_subnet | Remove

Buttons: Add new tag, Remove, Add new subnet.

At the bottom right, there are 'Cancel' and 'Create subnet' buttons. The 'Create subnet' button is highlighted with a red box.

2. Now click on the **Create Subnet**

Step 3: Create and attach an Internet Gateway

1. Internet Gateways → Create Internet Gateway

- Name tag : Enter **On_Premises_IGW**

2. Click on the **Create Internet Gateway**

PC > Internet gateways > Create internet gateway

Create internet gateway Info

An internet gateway is a virtual router that connects a VPC to the internet. To create a new internet gateway specify the name for the gateway below.

Internet gateway settings

Name tag
Creates a tag with a key of 'Name' and a value that you specify.

On_Premises_IGW

Tags - optional
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key **Value - optional**

Q Name X Q On_Premises_IGW X Remove

Add new tag

You can add 49 more tags.

Cancel Create internet gateway

- Now click on the **Actions** and select **Attach to VPC**.
 - Available VPCs : select **On_Premises_Network** from the list.
- Click on the **Attach Internet Gateway**

VPC > Internet gateways > Attach to VPC (igw-010fac61cb469297d)

Attach to VPC (igw-010fac61cb469297d) Info

VPC
Attach an internet gateway to a VPC to enable the VPC to communicate with the internet. Specify the VPC to attach below.

Available VPCs

Q vpc-04d4403d67a7a052e X

► AWS Command Line Interface command

Cancel Attach internet gateway

Step 4: Create a Public Route Table and associate it with the subnet

- Route Tables** → **Create route table** button.
 - Name tag : Enter **PublicRT**
 - VPC* : Select the **On_Premises_Network** from the list.
- Click on the **Create route table** button.
- Now click on **Route Tables** in the left panel and select the **PublicRT** from the list and go to the **Subnet associations** tab in below.
- Click on the **Edit subnet associations** button.
- Now select the subnet **Public_subnet** and click on the **Save associations** button.

bles > Create route table

Create route table Info

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

Route table settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.

PublicRT

VPC
The VPC to use for this route table.

vpc-04d4403d67a7a052e (On_Premises_Network)

Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Q Name X

Value - optional

Q PublicRT X Remove

Add new tag

You can add 49 more tags.

Cancel Create route table

rtb-055c25233d39c1457 > Edit subnet associations

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (1/1)

Filter subnet associations

☒ Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
☒ Public_subnet	subnet-0d3c11576459d88de	10.0.1.0/24	-

Selected subnets

subnet-0d3c11576459d88de / Public_subnet X

Cancel Save associations

Step 5: Add public Route in the Route table

1. Route Tables → PublicRT
2. Routes → Edit routes button.
3. Now click on the **Add route** button.
 - Destination : Enter **0.0.0.0/0**
 - Target : select **Internet Gateway** and then select the **Internet Gateway id**.

Edit routes

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	CreateRouteTable
0.0.0.0/0	Internet Gateway		No	CreateRoute

Add route

Cancel Preview Save changes

4. Click on the **Save changes** button.

Step 6: Create Security Group

1. EC2 → security groups → Create security group

- Security group name : Enter ***On_Premises_Router_SG***
- Description : Enter ***Security group for public Router***
- To add **SSH**,
 - Choose Type: **SSH**
 - Source: **Anywhere** (From ALL IP addresses accessible).
- For **All TCP**,
 - Click on **Add Security group rule** button.
 - Choose Type: **All TCP**
 - Source: **Custom** and Enter **30.0.0.0/16** in the textbox.
- For **IPv4 ICMP**,
 - Click on **Add Security group rule** button.
 - Choose Type: **All ICMP - IPv4**
 - Source: **Custom** and Enter **30.0.0.0/16** in the textbox.

VPC > Security Groups > sg-045c5da132a261cbd - On_Premises_Router_SG > Edit inbound rules

Edit inbound rules Info

Inbound rules control the incoming traffic that's allowed to reach the instance.

Security group rule ID	Type Info	Protocol Info	Port range Info	Source Info	Description - optional Info	
sgr-074af366bef9b7995	All ICMP - IPv4	ICMP	All	Custom	30.0.0.0/16	Delete
sgr-0bb7adb54607d6bb0	SSH	TCP	22	Custom	0.0.0.0/0	Delete
sgr-034170857abe612e2	All TCP	TCP	0 - 65535	Custom	30.0.0.0/16	Delete

Add rule

Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend restricting security group rules to allow access from known IP addresses only.

Step 7: Launch an EC2 instance

1. EC2 → Instances → Launch Instances
2. Enter Name as ***On_Premises_Router***
3. Choose an Amazon Machine Image (AMI): Search for **Amazon Linux 2023 kernel-6.1 AMI** in the search box and click on the **select** button.

Quick Start

Amazon Linux
aws

macOS
Mac

Ubuntu
ubuntu

Windows
Microsoft

Red Hat
Red Hat

SUSE Linux
SUSE

Debian
debian

[Browse more AMIs](#)
Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.1 AMI
ami-0a1235697f4afa8a4 (64-bit (x86), uefi-preferred) / ami-03e81965fd8e52909 (64-bit (Arm), uefi)
Virtualization: hvm ENA enabled: true Root device type: ebs

▼

Description

Amazon Linux 2023 (kernel-6.1) is a modern, general purpose Linux-based OS that comes with 5 years of long term support. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.

Amazon Linux 2023 AMI 2023.8.20250707.0 x86_64 HVM kernel-6.1

Architecture	Boot mode	AMI ID	Publish Date	Username	
64-bit (x86) ▼	uefi-preferred	ami-0a1235697f4afa8a4	2025-07-08	ec2-user	Verified provider

4. Choose an Instance Type: select **t3.micro**

▼ **Instance type** [Info](#) | [Get advice](#)

Instance type

t3.micro Free tier eligible
 Family: t3 2 vCPU 1 GiB Memory Current generation: true
 On-Demand Ubuntu Pro base pricing: 0.0139 USD per Hour
 On-Demand SUSE base pricing: 0.0104 USD per Hour
 On-Demand Linux base pricing: 0.0104 USD per Hour
 On-Demand RHEL base pricing: 0.0392 USD per Hour
 On-Demand Windows base pricing: 0.0196 USD per Hour

▼

[Additional costs apply for AMIs with pre-installed software](#)

☐ All generations [Compare instance types](#)

Key Pair : Choose **Create a new key Pair** from the dropdown list.

- Key pair name : Enter **Router-key**
- Key Pair Type: Select **RSA**
- Click on **Create key pair** button to download the key to your local machine.

5. Under **Network Settings**, click on **Edit** button.

- Network : Select **On_Premises_Network**
- Subnet : leave as default
- Auto-assign Public IP : Select **Enable**

▼ **Network settings** [Info](#)

VPC - required [Info](#)

vpc-04d4403d67a7a052e (On_Premises_Network) 10.0.0.0/16

Subnet [Info](#)

subnet-0d3c11576459d88de Public_subnet
 VPC: vpc-04d4403d67a7a052e Owner: 380106462512 Availability Zone: us-east-1d (use1-az6)
 Zone type: Availability Zone IP addresses available: 250 CIDR: 10.0.1.0/24

Auto-assign public IP [Info](#)

Enable

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☐ Create security group ☒ Select existing security group

Common security groups [Info](#)

Select security groups

On_Premises_Router_SG sg-0d5e7d58d4ffc7f8e X
 VPC: vpc-04d4403d67a7a052e

Security groups that you add or remove here will be added to or removed from all your network interfaces.

► **Advanced network configuration**

6. Configure Security Group:

- Firewall(security groups) : Select **existing security group**
- Select **On_Premises_Router_SG**

7. Click on **Launch Instance** button.

es > i-027f183eb1817aa1f

<

Instance summary for i-027f183eb1817aa1f (On_Premises_Router) [Info](#)

Updated less than a minute ago

Instance ID i-027f183eb1817aa1f	Public IPv4 address 35.153.135.244 open address	Private IPv4 addresses 10.0.1.102
IPv6 address -	Instance state Running	Public DNS -
Hostname type IP name: ip-10-0-1-102.ec2.internal	Private IP DNS name (IPv4 only) ip-10-0-1-102.ec2.internal	Elastic IP addresses -
Answer private resource DNS name -	Instance type t3.micro	AWS Compute Optimizer finding User: arn:aws:iam:471112956631:user/cloud_user is not authorized to perform: compute-optimizer:GetEnrollmentStatus on resource: * with an explicit deny in a service control policy
Auto-assigned IP address 35.153.135.244 [Public IP]	VPC ID vpc-00e159c57c01ce1b0 (On_Premises_Network)	

IS

Connect Instance state Actions

Step 8: Create AWS_Network VPC

1. VPC → Your VPC → Create VPC

- Select **VPC Only**

- Name tag : Enter **AWS_Network**
- IPv4 CIDR block : Enter **30.0.0.0/16**

2. Leave everything else as default and click on the **Create VPC**

VPC > Your VPCs > Create VPC

VPC settings

Resources to create [Info](#)
Create only the VPC resource or the VPC and other networking resources.

☒ VPC only ☐ VPC and more

Name tag - optional [Info](#)
Creates a tag with a key of 'Name' and a value that you specify.

AWS_Network

IPv4 CIDR block [Info](#)
☒ IPv4 CIDR manual input
☐ IPAM-allocated IPv4 CIDR block

IPv4 CIDR
30.0.0.0/16
CIDR block size must be between /16 and /28.

IPv6 CIDR block [Info](#)
☒ No IPv6 CIDR block
☐ IPAM-allocated IPv6 CIDR block
☐ Amazon-provided IPv6 CIDR block
☐ IPv6 CIDR owned by me

Tenancy [Info](#)
Default

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key	Value - optional	
Q Name	Q AWS_Network	Remove tag

[Add tag](#)

You can add 49 more tags

Cancel [Preview code](#) **Create VPC**

Step 9: Create a Private subnet, a Private Route Table and associate it with the subnet

1. Subnets → Create Subnets

- VPC ID : Select the **AWS_Network** VPC from the list.
- Subnet name : Enter **Private_subnet**
- Availability Zone : No preference
- IPv4 CIDR block : Enter **30.0.1.0/24**

Create subnet

VPC

VPC ID
Create subnets in this VPC

vpc-07213d9438ba1db99 (AWS_Network)

Associated VPC CIDRs

IPv4 CIDRs
30.0.0.0/16

Subnet settings

Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name
Create a tag with a key of 'Name' and a value that you specify.

Private_subnet
The name can be up to 256 characters long.

Availability Zone [Info](#)
Choose the zone in which your subnet will reside, or let Amazon choose one for you.

No preference

IPv4 VPC CIDR block [Info](#)
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

30.0.0.0/16

IPv4 subnet CIDR block

30.0.1.0/24 256 IPs

Tags - optional

Key	Value - optional	
Name	Private_subnet	Remove

[Add new tag](#)
You can add 49 more tags.

[Remove](#)

[Add new subnet](#)

[Cancel](#) [Create subnet](#)

2. Now click on the **Create Subnet**

Create a Private Route Table and associate it with the subnet

1. **Route Tables** → **Create route table**

- Name tag : Enter **PrivateRT**
- VPC* : Select the **AWS_Network** from the list.

Create route table [Info](#)

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

Route table settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.

PrivateRT

VPC
The VPC to use for this route table.

vpc-064e9512f611c1bef (AWS_Network)

Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key **Value - optional**

Q Name X Q PrivateRT X Remove

Add new tag

You can add 49 more tags.

Cancel Create route table

2. **Create route table** → **Route Tables** → **PrivateRT** from the list and go to the **Subnet associations** tab in below.
3. Click on the **Edit subnet associations** button.
4. Now select the subnet **Private_subnet** and click on the **Save associations** button.

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (1/1)

Q Filter subnet associations

☒	Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
☒	Private_subnet	subnet-0731de386e90860de	30.0.1.0/24	-	Main (rtb-0bc758bc5ea18f667)

Selected subnets

subnet-0731de386e90860de / Private_subnet X

Cancel Save associations

Now select the PrivateRT **RouteTable**, where VPC is mentioned as **AWS_Network**. And goto the **Route propagation** tab below and click on the **Edit route propagation** button.

Route tables (1/5) [Info](#) Last updated less than a minute ago [Actions](#) [Create route table](#)

Find route tables by attribute or tag

	Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
☐	-	rtb-0f495fdfbeabc16eb	-	-	Yes	vpc-0e2a827bbe
☐	-	rtb-0927c4f3349ca7bcc	-	-	Yes	vpc-00e159c57c
☐	PublicRT	rtb-0a091b77cae77e16f	subnet-035d12ec93d56b...	-	No	vpc-00e159c57c
☐	-	rtb-0bc758bc5ea18f667	-	-	Yes	vpc-064e9512f61
☒	PrivateRT	rtb-04168d21987bf44ec	subnet-0731de386e9086...	-	No	vpc-064e9512f61

rtb-04168d21987bf44ec / PrivateRT

[Details](#) [Routes](#) [Subnet associations](#) [Edge associations](#) [Route propagation](#) [Tags](#)

Details

Route table ID
[rtb-04168d21987bf44ec](#)

VPC
[vpc-064e9512f611c1bef](#) | [AWS_Network](#)

Main
☐ No

Owner ID
[471112956631](#)

Explicit subnet associations
[subnet-0731de386e90860de](#) / [Private_subnet](#)

Edge associations
-

Check the **Enable** checkbox and click on the **Save** button.

VPC > Route tables > [rtb-04168d21987bf44ec](#) > Edit route propagation

Edit route propagation

Route table basic details

Route table ID
[rtb-04168d21987bf44ec](#)

Edit route propagation

Virtual Private Gateway
[vgw-0885c0c1af1661abf](#) / [AWS_Network_VPG](#)

Propagation
☒ Enable

[Cancel](#) [Save](#)

Step 10: Create security group

1. EC2 → security groups → create security groups.

- Security group name : Enter **AWS_EC2_SG**
- Description : Enter **Security group for public Router**
 - To add **SSH**,
 - Choose Type: **SSH**
 - Source: **Anywhere** (From ALL IP addresses accessible).
 - For **All TCP**,
 - Click on **Add Security group rule** button.
 - Choose Type: **All TCP**
 - Source: **Custom** and Enter

- **10.0.0.0/16**
- in the textbox.
- For **IPv4 ICMP**,
 - Click on **Add Security group rule** button.
 - Choose Type: **All ICMP - IPv4**
 - Source: **Custom** and Enter
 - **10.0.0.0/16**
 - in the textbox.

Create security group [Info](#)
A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name [Info](#)

Name cannot be edited after creation.

Description [Info](#)

VPC [Info](#)

Inbound rules [Info](#)
This security group has no inbound rules.
[Add rule](#)

Outbound rules [Info](#)

Type	Protocol	Port range	Destination	Description - optional	
All TCP	TCP	0 - 65535	Custom	Q 10.0.0.0/16	Delete
SSH	TCP	22	Anywhere...	Q 0.0.0.0/0	Delete
All ICMP - IPv4	ICMP	All	Custom	Q 10.0.0.0/16	Delete

[Add rule](#)

Rules with destination of 0.0.0.0/0 or ::/0 allow your instances to send traffic to any IPv4 or IPv6 address. We recommend setting security group rules to be more restrictive and to only allow traffic to specific known IP addresses.

Tags - optional
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.
No tags associated with the resource.
[Add new tag](#)
You can add up to 50 more tags

Step 11: Launch an EC2 instance

1. **EC2 → Instances → Launch Instances**
2. Enter Name as **AWS_EC2**
3. **Choose an Amazon Machine Image (AMI):** Search for **Amazon Linux 2023 kernel-6.1 AMI** in the search box and click on the **select** button.

Quick Start

Amazon Linux
aws

macOS
Mac

Ubuntu
ubuntu

Windows
Microsoft

Red Hat
Red Hat

SUSE Linux
SUSE

Debian
debian

[Browse more AMIs](#)
Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.1 AMI
ami-0a1235697f4afa8a4 (64-bit (x86), uefi-preferred) / ami-03e81965fd8e52909 (64-bit (Arm), uefi)
Virtualization: hvm ENA enabled: true Root device type: ebs

▼

Description

Amazon Linux 2023 (kernel-6.1) is a modern, general purpose Linux-based OS that comes with 5 years of long term support. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.

Amazon Linux 2023 AMI 2023.8.20250707.0 x86_64 HVM kernel-6.1

Architecture	Boot mode	AMI ID	Publish Date	Username	
64-bit (x86) ▼	uefi-preferred	ami-0a1235697f4afa8a4	2025-07-08	ec2-user	Verified provider

4. Choose an Instance Type: select **t3.micro**

▼ **Instance type** [Info](#) | [Get advice](#)

Instance type

t3.micro
Family: t3 2 vCPU 1 GiB Memory Current generation: true
On-Demand Ubuntu Pro base pricing: 0.0139 USD per Hour
On-Demand SUSE base pricing: 0.0104 USD per Hour
On-Demand Linux base pricing: 0.0104 USD per Hour
On-Demand RHEL base pricing: 0.0392 USD per Hour
On-Demand Windows base pricing: 0.0196 USD per Hour

Free tier eligible
▼

[Additional costs apply for AMIs with pre-installed software](#)

☐ All generations

[Compare instance types](#)

Key Pair : Choose **Select key Pair** from the dropdown list.

- Key pair name : Enter **Route-key**

5. Under **Network Settings**, click on **Edit** button.

- Network : Select **AWS_Network**
- Subnet : leave as default
- Auto-assign Public IP : Select **Disable**

6. Configure Security Group:

- Firewall(security groups) : Select **existing security group**
- Select **AWS_EC2_SG**

▼ Key pair (login) Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

Router-key

Create new key pair

▼ Network settings Info

VPC - required Info

vpc-0679fe9536b5489f7 (AWS_Network)

30.0.0.0/16

Create new VPC

Subnet Info

subnet-0e4bd3d42a2b7b327

Private_subnet

VPC: vpc-0679fe9536b5489f7

Owner: 380106462512

Availability Zone: us-east-1c (use1-az4)

Zone type: Availability Zone

IP addresses available: 251

CIDR: 30.0.1.0/24

Create new subnet

Auto-assign public IP Info

Disable

Firewall (security groups) Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Create security group

Select existing security group

Common security groups Info

Select security groups

AWS_EC2_SG sg-037df4f249a30a87c

VPC: vpc-0679fe9536b5489f7

Compare security group rules

Security groups that you add or remove here will be added to or removed from all your network interfaces.

► Advanced network configuration

7. Click on **Launch Instance** button.

Find Instance by attribute or tag (case-sensitive)

All states

1

	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone
☐	On_Premises_...	i-027f183eb1817aa1f	Running	t3.micro	3/3 checks passec	View alarms +	us-east-1a
☐	AWS_EC2	i-0466471401a0e87ae	Running	t3.micro	3/3 checks passec	View alarms +	us-east-1a

AWS_EC2

i-0466471401a0e87ae

Running

t3.micro

3/3 checks passec

View alarms +

us-east-1a

i-0466471401a0e87ae (AWS_EC2)

▼ Instance summary Info

Instance ID

i-0466471401a0e87ae

IPv6 address

-

Hostname type

IP name: ip-30-0-1-227.ec2.internal

Public IPv4 address

-

Instance state

Running

Private IP DNS name (IPv4 only)

ip-30-0-1-227.ec2.internal

Private IPv4 addresses

30.0.1.227

Public DNS

-

Step 12: Create a Customer Gateway

1. **VPC→Customer Gateway→Virtual Private Network (VPN).**
2. Click on the **Create Customer Gateway**
 - Name : Enter **On_Premises_Network_Gateway**
 - IP Address : Enter **On-premises EC2 instance IPv4 Public IP (EC2 act as the Public Router)**
 - Leave everything else as default.

Create customer gateway [Info](#)

A customer gateway is a resource that you create in AWS that represents the customer gateway device in your on-premises network.

Details

Name tag - *optional*

Creates a tag with a key of 'Name' and a value that you specify.

On_Premises_Network_Gateway

Value must be 256 characters or less in length.

BGP ASN [Info](#)

The ASN of your customer gateway device.

65000

Value must be in 1 - 4294967294 range.

IP address [Info](#)

Specify the IP address for your customer gateway device's external interface.

35.153.135.244

Certificate ARN - *optional*

The ARN of a private certificate provisioned in AWS Certificate Manager (ACM).

Select certificate ARN

Click on the **Create Customer Gateway**

Customer gateways (1) Info								Actions	Create customer gateway
Find resource by attribute or tag									
	Name ↗	Customer gateway ID	State	BGP ASN	IP address	Type			
☐	On_Premises_Netwo...	cgw-0831c48238bfe99f7	Available	65000	35.153.135.244	ipsec.1			

Step 13: Create a Virtual Private Gateway

1. Now scroll down and select **Virtual Private Gateways** under **Virtual Private Network (VPN).**
2. Click on the **Create Virtual Private Gateway**
 - Name tag : Enter **AWS_Network_VPG**
 - ASN : Leave as Default

3. Now click on the **Create Virtual Private Gateway**

ys > Create virtual private gateway

Create virtual private gateway [Info](#)

A virtual private gateway is the VPN concentrator on the Amazon side of the site-to-site VPN connection.

Details

Name tag - optional
Enter a tag with a key and an optional value that you specify.

AWS_Network_VPG

Value must be 256 characters or less in length.

Autonomous System Number (ASN)

☒ Amazon default ASN
☐ Custom ASN

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs. Name tag helps you track your resources more easily. We recommend adding Name tag.

Key

Value - optional

You can add up to 49 more tags.

4. Now the state will be **detached** for the Virtual Private gateway that you have created.

Virtual private gateways (1) [Info](#)

	Name 🔗	Virtual private gateway ID	State	VPC attachment state	Type	VPC	Amazon ASN
☐	AWS_Network_VPG	vgw-Od17640723082c41c	Pending	Detached	ipsec.1	-	64512

5. You need to attach the VPG to the VPC

6. Select the **Virtual Private gateway** that you just created (**wait till the state changes to Available**), click on the **Actions** button and select **Attach to VPC**.

- VPC : Select **AWS_Network** from the list.
- Now click on the **Attach to VPC**

ways > vgw-Od17640723082c41c > Attach to VPC

Attach to VPC [Info](#)

Attach the virtual private gateway to this VPC.

Details

Virtual private gateway ID
 vgw-Od17640723082c41c

Available VPCs

vpc-0679fe9536b5489f7 / AWS_Network

Now select the PrivateRT **RouteTable**, where VPC is mentioned as **AWS_Network**. And goto the **Route propagation** tab below and click on the **Edit route propagation** button.

Route tables (1/5) [Info](#) Last updated less than a minute ago [Actions](#) [Create route table](#)

☐	Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
☐	-	rtb-0f495fdfbeabc16eb	-	-	Yes	vpc-0e2a827bbe
☐	-	rtb-0927c4f3349ca7bcc	-	-	Yes	vpc-00e159c57cc
☐	PublicRT	rtb-0a091b77cae77e16f	subnet-035d12ec93d56b...	-	No	vpc-00e159c57cc
☐	-	rtb-0bc758bc5ea18f667	-	-	Yes	vpc-064e9512f61
☒	PrivateRT	rtb-04168d21987bf44ec	subnet-0731de386e9086...	-	No	vpc-064e9512f61

rtb-04168d21987bf44ec / PrivateRT

[Details](#) [Routes](#) [Subnet associations](#) [Edge associations](#) [Route propagation](#) [Tags](#)

Details

Route table ID rtb-04168d21987bf44ec	Main ☐ No	Explicit subnet associations subnet-0731de386e90860de / Private_subnet	Edge associations -
VPC vpc-064e9512f61c1bef AWS_Network	Owner ID 471112956631		

Check the **Enable** checkbox and click on the **Save** button.

[VPC](#) > [Route tables](#) > [rtb-04168d21987bf44ec](#) > Edit route propagation

Edit route propagation

Route table basic details

Route table ID
[rtb-04168d21987bf44ec](#)

Edit route propagation

Virtual Private Gateway
[vgw-0885c0c1af1661abf / AWS_Network_VPG](#)

Propagation
☒ Enable

[Cancel](#) [Save](#)

Step 14: Create a Site-to-Site VPN connection

1. VPC→Site-to-Site VPN Connections→Virtual Private

Network(VPN)-->Create VPN Connection

- Name tag : Enter **AWS_On_Premises_Connection**
- Virtual Private Gateway* : Select **AWS_Network_VPG** from the list.
- Customer Gateway ID* : Select **On_Premises_Network** from the list.
- Routing Options : Select **Static**
- Static IP Prefixes : Enter **10.0.0.0/16** in the textbox (Your On-Premises VPC CIDR)

Create VPN connection [Info](#)

Select the resources and additional configuration options that you want to use for the site-to-site VPN connection.

Details

Name tag - optional

Creates a tag with a key of 'Name' and a value that you specify.

AWS_On_Premises_Connection

Value must be 256 characters or less in length.

Target gateway type [Info](#)

- ☒ Virtual private gateway
☐ Transit gateway
☐ Not associated

Virtual private gateway

vgw-0d17640723082c41c

Customer gateway [Info](#)

- ☒ Existing
☐ New

Customer gateway ID

cgw-0309db3e5c8b7d669

Routing options [Info](#)

- ☐ Dynamic (requires BGP)
☒ Static

Static IP prefixes [Info](#)

10.0.0.0/16

Tunnel Options :

- Leave as default.

3. Click on the **Create VPN Connection**. It will take 3 to 4 Minutes to set up the VPN Connection.

VPN connections (1/1) Info							Actions Download configuration Create VPN connection	
Find resource by attribute or tag								
Name	VPN ID	State	Virtual private gateway	Transit gateway	Customer gateway	Customer gateway IP		
AWS_On_Premises_...	vpn-07462872d85be6c86	Available	vgw-0cfa8213c74b17e21	-	cgw-01ab6f9edc074a86b	3.110.30.245		

4. Select the VPN Connection and on top, click on the **Download the configuration**

- Vendor : Select **Openswan**
- Platform : Select **Openswan**
- Software : Leave default.

Download configuration

×

Download a sample configuration based on your customer gateway. Note that this is a sample only, and that it will require modification for using Advanced Algorithms, Certificates, and/IPv6.

Vendor
The manufacturer of the customer gateway device (for example, Cisco Systems, Inc).

Openswan

Platform
The class of the customer gateway device (for example, J-Series).

Openswan

Software
The operating system running on the customer gateway device (for example, ScreenOS).

Openswan 2.6.38+

IKE version
The IKE version you are using for your VPN connection.

ikev1

Include sample type - optional
☐ Enable

Sample type
The default sample type compatibility mode includes all options. The recommended mode restricts options to only the most secure settings (IKEv2, etc.).

Select sample type

Cancel

Download

5. Click on the **Download** button and the configuration file will be downloaded to your local machine and note down the file name.

Step 15: Configure On-Premises Router

1. You need to SSH into the EC2 instance (**On_Premises_Router**) that you have created in the **On_Premises_Network**. This EC2 will act as the Public Router of On-Premises Network.
2. SSH to EC2 instance

```
ssh -i "Router-key.pem" ec2-user@35.153.135.244
```

Switch to root user :

```
sudo -s
```

3. Install Openswan

```
sudo dnf install libreswan -y
```

4. Next make sure the last line in `/etc/ipsec.conf` is not commented. (NO # in the beginning)

```
nano /etc/ipsec.conf
```

- Scroll to the end and make sure the last line include `/etc/ipsec.d/*.conf` has no hash (#) in the beginning.

```
# if it exists, include system wide crypto-policy def
# include /etc/crypto-policies/back-ends/libreswan.co
# It is best to add your IPsec connections as separat
include /etc/ipsec.d/*.conf
```

- Click **[Control] + X** or **[Ctrl] + X** to exit the file.

5. Update `/etc/sysctl.conf` file

```
nano /etc/sysctl.conf
```

- Add the below 3 lines in end of this file with no hash (#) in the beginning and add each in new lines

```
net.ipv4.ip_forward = 1
net.ipv4.conf.all.accept_redirects = 0
net.ipv4.conf.all.send_redirects = 0
```

```
# sysctl settings are defined through files in
# /usr/lib/sysctl.d/, /run/sysctl.d/, and /etc/sysctl.d/.
#
# Vendors settings live in /usr/lib/sysctl.d/.
# To override a whole file, create a new file with the same in
# /etc/sysctl.d/ and put new settings there. To override
# only specific settings, add a file with a lexically later
# name in /etc/sysctl.d/ and put new settings there.
#
# For more information, see sysctl.conf(5) and sysctl.d(5).
net.ipv4.ip_forward = 1
net.ipv4.conf.all.accept_redirects = 0
net.ipv4.conf.all.send_redirects = 0
```

- Click **[Control] + X** or **[Ctrl] + X** to exit the file.
- Save modified buffer (Answering "No" will DISCARD changes.) : Enter **Y**
- File Name to Write: /etc/sysctl.conf : Hit **[Enter]** Key

6. Restart Network Service

```
sudo systemctl restart systemd-networkd
```

7. Next we need to configure IPsec and pre-shared keys in Openswan.
8. For that you have **downloaded a configuration file** from the Site-to-site VPN page to your **local system**. Open that file in your text editor.
9. Create or open **/etc/ipsec.d/aws.conf** file

```
vi /etc/ipsec.d/aws.conf
```

- In the Configuration file, look for **point number 4** under **IPSEC Tunnel #1** and copy the entire code below **point number 4**.

IPSEC Tunnel #1

This configuration assumes that you already have a default `openswan` installation in place on the Amazon Linux operating system (but may also work with other `distros` as well)

1) Open `/etc/sysctl.conf` and ensure that its values match the following:

```
net.ipv4.ip_forward = 1
net.ipv4.conf.default.rp_filter = 0
net.ipv4.conf.default.accept_source_route = 0
```

2) Apply the changes in step 1 by executing the command `'sysctl -p'`

3) Open `/etc/ipsec.conf` and look for the line below. Ensure that the `#` in front of the line has been removed, then save and exit the file.

```
#include /etc/ipsec.d/*.conf
```

4) Create a new file at `/etc/ipsec.d/awes.conf` if doesn't already exist, and then open it. Append the following configuration to the end in the file:

`#leftsubnet=` is the local network behind your `openswan` server, and you will need to replace the `<LOCAL NETWORK>` below with this value (don't include the brackets). If you have `0.0.0.0/0` instead.

`#rightsubnet=` is the remote network on the other side of your VPN tunnel that you wish to have connectivity with, and you will need to replace `<REMOTE NETWORK>` with this value (don't include the brackets).

```
conn Tunnel1
    authby=secret
    auto=start
    left=%defaultroute
    leftid=35.153.135.244
    right=34.192.156.108
    type=tunnel
    ikelifetime=8h
    keylife=1h
    phase2alg=aes128-sha1;modp1024
    ike=aes128-sha1;modp1024
    auth=esp
    keyingtries=%forever
    keyexchange=ike
    leftsubnet=<LOCAL NETWORK>
    rightsubnet=<REMOTE NETWORK>
    dpddelay=10
    dpdtimeout=30
    dpdaction=restart_by_peer
```

5) Create a new file at `/etc/ipsec.d/awes.secrets` if it doesn't already exist, and append this line to the file (be mindful of the spacing!):

```
35.153.135.244 34.192.156.108: PSK "5.a8hhH8HrevtbEvd9D4XZrLEZSLuPPq"
```

- Press `"i"` button in your keyboard to edit the file you have created.
- Paste the code in the file we opened.
- **Remove** the line `auth=esp` from the file (else the connection won't work)

```
conn Tunnel1
    authby=secret
    auto=start
    left=%defaultroute
    leftid=35.153.135.244
    right=34.192.156.108
    type=tunnel
    ikelifetime=8h
    keylife=1h
    phase2alg=aes128-sha1;modp2048
    ike=aes128-sha1;modp2048
    keyingtries=%forever
    keyexchange=ike
    leftsubnet=10.0.0.0/16
    rightsubnet=30.0.0.0/16
    dpddelay=10
    dpdtimeout=30
    dpdaction=restart_by_peer
```

- Update
 - `phase2alg=aes128-sha1;modp1024`
 - `ike=aes128-sha1;modp1024`
- To
 - `phase2alg=aes128-sha1;modp2048`
 - `ike=aes128-sha1;modp2048`
- Update `leftsubnet` and `rightsubnet`
 - `leftsubnet=` Enter `10.0.0.0/16` (On-premises VPC CIDR)
 - `rightsubnet=` Enter `30.0.0.0/16` (AWS VPC CIDR)

```
conn Tunnel1
    authby=secret
    auto=start
    left=%defaultroute
    leftid=35.153.135.244
    right=34.192.156.108
    type=tunnel
    ikelifetime=8h
    keylife=1h
    phase2alg=aes128-sha1;modp2048
    ike=aes128-sha1;modp2048
    keyingtries=%forever
    keyexchange=ike
    leftsubnet=10.0.0.0/16
    rightsubnet=30.0.0.0/16
    dpddelay=10
    dpdtimeout=30
    dpdaction=restart_by_peer
```

- Click **[Esc]** button in the keyboard to exit the editing mode.
- Now type `:wq` and hit **[Enter]** Key to save the file.

10. Create or open `/etc/ipsec.d/aws.secrets` file

```
vi /etc/ipsec.d/aws.secrets
```

- In the Configuration file, look for **point number 5** under **IPSEC Tunnel #1** and copy the entire code below.

```

1) Open /etc/sysctl.conf and ensure that its values match the following:
net.ipv4.ip_forward = 1
net.ipv4.conf.default.rp_filter = 0
net.ipv4.conf.default.accept_source_route = 0

2) Apply the changes in step 1 by executing the command 'sysctl -p'

3) Open /etc/ipsec.conf and look for the line below. Ensure that the # in front of the line has been removed, then save and
#include /etc/ipsec.d/*.conf

4) Create a new file at /etc/ipsec.d/aws.conf if doesn't already exist, and then open it. Append the following configuration
#leftsubnet= is the local network behind your openswan server, and you will need to replace the <LOCAL NETWORK> below with
0.0.0.0/0 instead.
#rightsubnet= is the remote network on the other side of your VPN tunnel that you wish to have connectivity with, and you w

conn Tunnel1
    authby=secret
    auto=start
    left=%defaultroute
    leftid=35.153.135.244
    right=34.192.156.108
    type=tunnel
    ikelifetime=8h
    keylife=1h
    phase2alg=aes128-sha1:modp1024
    ike=aes128-sha1:modp1024
    auth=esp
    keyingtries=%forever
    keyexchange=ike
    leftsubnet=<LOCAL NETWORK>
    rightsubnet=<REMOTE NETWORK>
    dpddelay=10
    dpdtimeout=30
    dpdaction=restart_by_peer

5) Create a new file at /etc/ipsec.d/aws.secrets if it doesn't already exist, and append this line to the file (be mindful o
35.153.135.244 34.192.156.108: PSK "5.a8HhH8HreytbEvd9D4XZrLEZSLuPPq"

```

- Press " i " button in your keyboard to edit the file you have created.
- Paste the secret key in the file you opened.

```
vi /etc/ipsec.d/aws.secrets
```

```
35.153.135.244 34.192.156.108: PSK "5.a8HhH8HreytbEvd9D4XZrLEZSLuPPq"
```

- Click **[Esc]** button in the keyboard to exit the editing mode.
- Now type **:wq** and hit **[Enter]** Key to save the file.

11. In case if your EC2 Session gets timed out. please follow the steps to SSH into EC2 Instance again.

- Switch to root user :

```
sudo -s
```

12. Start IPsec service

```
systemctl start ipsec
```

13. Check the status of IPsec

```
systemctl status ipsec
```

```
[root@ip-10-0-1-12 ec2-user]# sudo -s
[root@ip-10-0-1-12 ec2-user]# systemctl start ipsec
[root@ip-10-0-1-12 ec2-user]# systemctl status ipsec
● ipsec.service - Internet Key Exchange (IKE) Protocol Daemon for IPsec
   Loaded: loaded (/usr/lib/systemd/system/ipsec.service; disabled; vendor preset: disabled)
   Active: active (running) since Tue 2025-06-03 06:37:30 UTC; 4s ago
     Docs: man:ipsec(8)
           man:pluto(8)
           man:ipsec.conf(5)
   Process: 4491 ExecStartPre=/usr/sbin/ipsec --checknflag (code=exited, status=0/SUCCESS)
   Process: 4484 ExecStartPre=/usr/sbin/ipsec --checknss (code=exited, status=0/SUCCESS)
   Process: 3968 ExecStartPre=/usr/libexec/ipsec/_stackmanager start (code=exited, status=0/SUCCESS)
   Process: 3967 ExecStartPre=/usr/libexec/ipsec/addconn --config /etc/ipsec.conf --checkconfig (code=exited, status=0/SUCCESS)
   Main PID: 4509 (pluto)
   Status: "Startup completed."
   CGroup: /system.slice/ipsec.service
           └─4509 /usr/libexec/ipsec/pluto --leak-detective --config /etc/ipsec.conf --nofork
```

Step 16: Test the connectivity between two Networks

1. Ping OnPrem EC2 to AWS Private EC2

- **ping** <Private IPv4 Address of On_Premises_Router>
- Example : **ping 30.0.1.227**

```
[root@ip-10-0-1-102 ec2-user]# ping 30.0.1.227
PING 30.0.1.227 (30.0.1.227) 56(84) bytes of data.
64 bytes from 30.0.1.227: icmp_seq=1 ttl=127 time=1.81 ms
64 bytes from 30.0.1.227: icmp_seq=2 ttl=127 time=1.77 ms
64 bytes from 30.0.1.227: icmp_seq=3 ttl=127 time=1.71 ms
64 bytes from 30.0.1.227: icmp_seq=4 ttl=127 time=1.94 ms
^C
```

2. Now scroll down and select **Site-to-Site VPN Connection** under **Virtual Private Network(VPC)**.
3. Check the **Tunnel Details** and you will be able to see that **Tunnel 1** is **UP**. Tunnel 2 is Down because in Openswan only one tunnel can be configured and we only used

Tunnel 1.

AWS_On_Premises_...

vpn-00ec10b457ecfea86

Available

vgw-0885c0c1af1661abf

-

cgw-0885c0c1af1661abf

VPN connection vpn-00ec10b457ecfea86 / AWS_On_Premises_Connection

⚠ This VPN connection is not using both tunnels. This mode of operation is not highly available and we strongly recommend you configure your second tunnel.

Tunnel state

Tunnel number ▾	Outside IP address ▾	Inside IPv4 CIDR ▾	Inside IPv6 CIDR ▾	Status ▾	Provisioning status ▾	Last status change
Tunnel 1	34.192.156.108	169.254.248.60/30	-	Up	Available	September 20, 2025, 13:09:32
Tunnel 2	34.206.225.123	169.254.177.36/30	-	Down	Available	September 20, 2025, 12:40:03

Step 17: Delete AWS Resources

1. Terminate both EC2 instances (On_Premises_Router and AWS_EC2).
2. Delete the Security Groups created for the lab (On_Premises_Router_SG and AWS_EC2_SG).
3. Detach and delete the Internet Gateway from the On_Premises_Network VPC.
4. Delete the subnets (Public_subnet and Private_subnet).
5. Delete the Route Tables (PublicRT and PrivateRT).
6. Delete the Customer Gateway and the Virtual Private Gateway.
7. Delete the Site-to-Site VPN Connection.
8. Finally, delete both VPCs (On_Premises_Network and AWS_Network).